PROPOSED (S-2 MC CIRCUIT CANDIDATE) FPGA: autonomous local rails_conditioned dual watchdogs, permission/OPI/STO latches and 12-motor isR buffers.
MC_RAILS_EN uses qualified destination—not FPGA_CRESET_N, not MC_RAILS_OK_5V; CDONE qualification uses delayed FPGA_USER_RESET_N; timing is unqualified.
Legacy progress ball is proposed as FPGA_PROGRESS_M_X2 is the independent RAW_INHIBIT_N input. Reserved enable balls now connect to EN_N buffers; vendor ITI, CNC1 pulse generation and remaining GPIO allocation are open.
Power: 3V3 and LOGIC_GND are shared with the MC board; VMOT and 5V remain on their own powered rail; those restraints remain separate unfinished circuits.
Native schematic ERC is distinct from PCB DRC; vendor FPGA fit and measurements, NOT FOR MANUFACTURE.

## fpga_core

CORE_3V2
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
EN0_N_FPGA
EN10_N_FPGA
EN11_N_FPGA
EN1_N_FPGA
EN2_N_FPGA
EN3_N_FPGA
EN4_N_FPGA
EN5_N_FPGA
EN6_N_FPGA
EN7_N_FPGA
EN8_N_FPGA
EN9_N_FPGA
FPGA_CRESET_N
FPGA_ENDPOINT_READY
FPGA_MOTION_REQUEST
FPGA_PROGRESS_M
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
RAW_INHIBIT_N
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TX_PERMIT

File: fpga_core.kicad_sch

## mc_local_power

CORE_3V2
FPGA_CRESET_N
IO_3V3
LOGIC_GND

File: mc_local_power.kicad_sch

## watchdog_conditioning

FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_PROGRESS_M
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
WD_M_OK
WD_M_SETS_BUF
WD_O_OK
WD_O_SETS_BUF

File: watchdog_conditioning.kicad_sch

## mc_permission

BANK_B_BUFFER_ENABLE_QUALIFIED
FPGA_CRESET_N
FPGA_ENDPOINT_READY
FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_USER_RESET_N
IO_3V3
LOGIC_GND
RAW_INHIBIT_N
TX_PERMIT
WD_M_OK
WD_M_SETS_BUF
WD_O_OK
WD_O_SETS_BUF

File: mc_permission.kicad_sch

## mc_motor_buffers

BANK_B_BUFFER_ENABLE_QUALIFIED
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
EN0_N_FPGA
EN10_N_FPGA
EN11_N_FPGA
EN1_N_FPGA
EN2_N_FPGA
EN3_N_FPGA
EN4_N_FPGA
EN5_N_FPGA
EN6_N_FPGA
EN7_N_FPGA
EN8_N_FPGA
EN9_N_FPGA
IO_3V3
LOGIC_GND
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TX_PERMIT

File: mc_motor_buffers.kicad_sch

Sheet: /
File: motion_card.kicad_sch
**Title: Motion card circuit candidate**
Size: A0 | Date: 2026-09-30 | **Rev: r0.A-proposed-S2**
KiCad 10.0.1 | Id: 1/1

# T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12-CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20-PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCCA=1.2V. No 5V-tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball: bulk per physical supply group. Decoupling is a layout-study allowance, not a PDN proof.
Active SPI x1 strap=1111: CS/TEST high. Flash bus reserved from user RTL. CCK dampling336 is provisional.
CDONE is open-drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not T1.

U1001
T20F256A / CANDIDATE PIN ALLOCATION

| | | | |
|---|---|---|---|
| FPGA_CDONE | 1 | CDONE | |
| FPGA_CRESET_N | 2 | CRESET_N | |
| FPGA_FLASH_CS_N | 3 | GPIOL_00_SS_N | |
| FPGA_FLASH_CCK_SOURCE | 4 | GPIOL_01_CCK | |
| unUSED_FPGA | 5 | GPIOL_02 | |
| FPGA_FLT0 | 6 | GPIOL_04_D0 | |
| | 7 | GPIOL_05_D1 | |
| FPGA_FLASH_MISO | 8 | GPIOL_06_D2 | |
| FPGA_FLASH_MOSI | 9 | GPIOL_09_D3 | |
| FPGA_CRAM_TCK | 10 | TCK | |
| FPGA_CRAM_TDI | 11 | TDI | |
| FPGA_CRAM_TDO | 12 | TDO | |
| FPGA_CRAM_TMS | 13 | TMS | |
| unUSED_FPGA | 14 | GPIOL_12 | |
| | 15 | GPIOL_13_CDI2 | |
| unUSED_FPGA | 16 | GPIOL_14 | |
| unUSED_FPGA | 17 | GPIOL_15 | |
| unUSED_FPGA | 18 | GPIOL_16 | |
| unUSED_FPGA | 19 | GPIOL_18 | |
| unUSED_FPGA | 20 | GPIOL_19 | |
| | 21 | GPIOL_20_CFM_D_CDI4 | |
| | 22 | GPIOL_21_CFM_D_CDI5 | |
| unUSED_FPGA | 23 | GPIOL_22_CFM_D | |
| unUSED_FPGA | 24 | GPIOL_23_CFM_D | |
| FPGA_CK_SDM | 25 | GPIOL_24_CLK0 | |
| unUSED_FPGA | 26 | GPIOL_25_CLK1 | |
| unUSED_FPGA | 27 | GPIOL_26_CLK2 | |
| unUSED_FPGA | 28 | GPIOL_27_CLK3 | |
| unUSED_FPGA | 29 | GPIOL_28_CLK4 | |
| unUSED_FPGA | 30 | GPIOL_29_CLK5 | |
| unUSED_FPGA | 31 | GPIOL_30_CLK6 | |
| | 32 | GPIOL_31_CLK7_CDI6 | |
| | 33 | GPIOL_32_CFM_A_CDI7 | |
| unUSED_FPGA | 34 | GPIOL_33_CFM_D | |
| unUSED_FPGA | 35 | GPIOL_34_CFM_D | |
| unUSED_FPGA | 36 | GPIOL_35_CFM_T | |
| unUSED_FPGA | 37 | GPIOL_36 | |
| unUSED_FPGA | 38 | GPIOL_37 | |
| unUSED_FPGA | 39 | GPIOL_39 | |
| | 40 | GPIOL_40_CDI8 | |
| | 41 | GPIOL_41_CDI9 | |
| unUSED_FPGA | 42 | GPIOL_42 | |
| unUSED_FPGA | 43 | GPIOL_43 | |
| unUSED_FPGA | 44 | GPIOL_44 | |
| unUSED_FPGA | 45 | GPIOL_45 | |
| HOST_SPI_CLK | 46 | GPIOR_00_CCLK | |
| HOST_SPI_CS_N | 47 | GPIOR_01_nCSO | |
| HOST_SPI_MOSI | 48 | GPIOR_02_MOSI_D0 | |
| HOST_SPI_MISO | 49 | GPIOR_03_MISO_D1 | |
| FPGA_USER_RESET_N | 50 | GPIOR_04_D2 | |
| SDM_STATUS_FPGA | 51 | GPIOR_05_D3 | |
| PROTECTIVE_STATUS_FPGA | 52 | GPIOR_06_CSI2 | |
| SPIMOLE_FAULT_FPGA | 53 | GPIOR_07_CSI3 | |
| SPIMOLE_BOARD_FPGA | 54 | GPIOR_08 | |
| TOOLSETTER_FPGA | 55 | GPIOR_09_CSO1 | |
| CYCLE_START_FPGA | 56 | GPIOR_10_CSO2 | |
| LUBE_STATUS_FPGA | 57 | GPIOR_11_CSO3 | |
| | 58 | GPIOR_12_DQ0 | |
| | 59 | GPIOR_13_DQ1 | |
| | 60 | GPIOR_14_DQ2 | |
| | 61 | GPIOR_15_DQ3 | |
| | 62 | GPIOR_17_PLL4 | |
| | 63 | GPIOR_18_PLL4 | |
| | 64 | GPIOR_19_PLL4 | |
| | 65 | GPIOR_20_PLL4 | |
| | 66 | GPIOR_21_PLL4 | |
| | 67 | GPIOR_22_PLL4 | |
| | 68 | GPIOR_23_PLL4 | |
| | 69 | GPIOR_24_PLL4 | |
| | 70 | GPIOR_25_PLL4 | |
| | 71 | GPIOR_26_PLL4 | |
| | 72 | GPIOR_27_PLL4 | |
| | 73 | GPIOR_28_PLL4 | |
| | 74 | GPIOR_29_PLL4 | |
| | 75 | GPIOR_30_PLL4 | |
| | 76 | GPIOR_31_PLL4 | |
| | 77 | GPIOR_32_PLL4 | |
| | 78 | GPIOR_33_PLL4 | |
| | 79 | GPIOR_34_PLL4 | |
| | 80 | GPIOR_35_PLL4 | |
| | 81 | GPIOR_36_PLL4 | |
| | 82 | GPIOR_37_PLL4 | |
| | 83 | GPIOR_38_PLL4 | |
| | 84 | GPIOR_39_PLL4 | |
| | 85 | GPIOR_40_PLL4 | |
| | 86 | GPIOR_41_PLL4 | |
| | 87 | GPIOR_42_PLL4 | |
| | 88 | GPIOR_43_PLL4 | |
| | 89 | GPIOR_44_PLL4 | |
| | 90 | GPIOR_45_PLL4 | |
| | 91 | VCCIO4_1V8 | |
| | 92 | VCCIO4_3V3 | |
| | 93 | VCCIO3B_3V3 | |
| | 94 | VCCIO3A_3V3 | |
| | 95 | VCCIO3A_3V3 | |
| | 96 | VCCIO3A_3V3 | |
| | 97 | VCCIO3B_3V3 | |
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| | 253 | VCCIO3B_3V3 | |
| | 254 | VCCIO3B_3V3 | |
| | 255 | VCCIO3B_3V3 | |
| | 256 | VCCIO3B_3V3 | |

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE-qualified user-logic reset: R22/C39 provide a candidate post-CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User-logic reset release threshold, RC tolerance, discharge time, !USER handoff and slack/up/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary; actual keyed connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate reset cycle, valid rails and machine outputs inhibited. No hot-plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bifobream, qualification or fabrication files.

U1002
W25Q32JVSIQ / 4 x 5M

| | | | |
|---|---|---|---|
| FPGA_FLASH_CS_N | 1 | /CS | |
| FPGA_FLASH_CCK | 6 | CLK | |
| FPGA_FLASH_MOSI | 5 | DI_IO0 | |
| | 8 | VCC | 3V3 |
| FPGA_FLASH_MISO | 2 | DO_IO1 | |
| FPGA_FLASH_IO2 | 3 | WP_N_IO2 | |
| FPGA_FLASH_IO3 | 7 | HOLD_N_IO3 | |
| | 4 | GND | 0VDC_GND |

U1003
XC7K160T-2FBG484I / 4 x 5M

| | | | |
|---|---|---|---|
| 3V3 | 1 | VCCBATT_0 | |
| | 2 | GND | 0VDC_GND |
| | 3 | VCCAUX | 3V3 |
| | 4 | VCCO_0 | 3V3 |
| | 5 | VCCO_14 | 3V3 |
| | 6 | VCCO_15 | 3V3 |
| | 7 | VCCO_16 | 3V3 |
| | 8 | VCCO_17 | 3V3 |
| | 9 | VCCO_18 | 3V3 |
| | 10 | VCCO_19 | 3V3 |
| | 11 | VCCO_20 | 3V3 |
| | 12 | VCCO_21 | 3V3 |
| | 13 | VCCO_22 | 3V3 |
| | 14 | VCCO_23 | 3V3 |
| | 15 | VCCO_24 | 3V3 |
| | 16 | VCCO_25 | 3V3 |
| | 17 | VCCO_26 | 3V3 |
| | 18 | VCCO_27 | 3V3 |
| | 19 | VCCO_28 | 3V3 |
| | 20 | VCCO_29 | 3V3 |
| | 21 | VCCO_34 | 3V3 |
| | 22 | VCCO_35 | 3V3 |
| | 23 | VCCO_36 | 3V3 |
| | 24 | VCCO_37 | 3V3 |
| | 25 | VCCO_38 | 3V3 |
| | 26 | VCCO_39 | 3V3 |
| | 27 | VCCO_40 | 3V3 |
| | 28 | VCCO_41 | 3V3 |
| | 29 | VCCO_42 | 3V3 |
| | 30 | VCCO_43 | 3V3 |
| | 31 | VCCO_44 | 3V3 |
| | 32 | VCCO_45 | 3V3 |
| | 33 | VCCO_46 | 3V3 |
| | 34 | VCCO_47 | 3V3 |
| | 35 | VCCO_48 | 3V3 |
| | 36 | VCCO_49 | 3V3 |
| | 37 | VCCO_50 | 3V3 |
| | 38 | VCCO_51 | 3V3 |
| | 39 | VCCO_52 | 3V3 |
| | 40 | VCCO_53 | 3V3 |
| | 41 | VCCO_54 | 3V3 |
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| | 43 | VCCO_56 | 3V3 |
| | 44 | VCCO_57 | 3V3 |
| | 45 | VCCO_58 | 3V3 |
| | 46 | VCCO_59 | 3V3 |
| | 47 | VCCO_60 | 3V3 |
| | 48 | VCCO_61 | 3V3 |
| | 49 | VCCO_62 | 3V3 |
| | 50 | VCCO_63 | 3V3 |
| | 51 | VCCO_64 | 3V3 |
| | 52 | VCCO_65 | 3V3 |
| | 53 | VCCO_66 | 3V3 |
| | 54 | VCCO_67 | 3V3 |
| | 55 | VCCO_68 | 3V3 |
| | 56 | VCCO_69 | 3V3 |
| | 57 | VCCO_70 | 3V3 |
| | 58 | VCCO_71 | 3V3 |
| | 59 | VCCO_72 | 3V3 |
| | 60 | VCCO_73 | 3V3 |
| | 61 | VCCO_74 | 3V3 |
| | 62 | VCCO_75 | 3V3 |
| | 63 | VCCO_76 | 3V3 |
| | 64 | VCCO_77 | 3V3 |
| | 65 | VCCO_78 | 3V3 |
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| | 67 | VCCO_80 | 3V3 |
| | 68 | VCCO_81 | 3V3 |
| | 69 | VCCO_82 | 3V3 |
| | 70 | VCCO_83 | 3V3 |
| | 71 | VCCO_84 | 3V3 |
| | 72 | VCCO_85 | 3V3 |
| | 73 | VCCO_86 | 3V3 |
| | 74 | VCCO_87 | 3V3 |
| | 75 | VCCO_88 | 3V3 |
| | 76 | VCCO_89 | 3V3 |
| | 77 | VCCO_90 | 3V3 |
| | 78 | VCCO_91 | 3V3 |
| | 79 | VCCO_92 | 3V3 |
| | 80 | VCCO_93 | 3V3 |
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| | 87 | VCCO_100 | 3V3 |
| | 88 | VCCO_101 | 3V3 |
| | 89 | VCCO_102 | 3V3 |
| | 90 | VCCO_103 | 3V3 |
| | 91 | VCCO_104 | 3V3 |
| | 92 | VCCO_105 | 3V3 |
| | 93 | VCCO_106 | 3V3 |
| | 94 | VCCO_107 | 3V3 |
| | 95 | VCCO_108 | 3V3 |
| | 96 | VCCO_109 | 3V3 |
| | 97 | VCCO_110 | 3V3 |
| | 98 | VCCO_111 | 3V3 |
| | 99 | VCCO_112 | 3V3 |
| | 100 | VCCO_113 | 3V3 |
| | 101 | VCCO_114 | 3V3 |
| | 102 | VCCO_115 | 3V3 |
| | 103 | VCCO_116 | 3V3 |
| | 104 | VCCO_117 | 3V3 |
| | 105 | VCCO_118 | 3V3 |
| | 106 | VCCO_119 | 3V3 |
| | 107 | VCCO_120 | 3V3 |
| | 108 | VCCO_121 | 3V3 |
| | 109 | VCCO_122 | 3V3 |
| | 110 | VCCO_123 | 3V3 |
| | 111 | VCCO_124 | 3V3 |
| | 112 | VCCO_125 | 3V3 |
| | 113 | VCCO_126 | 3V3 |
| | 114 | VCCO_127 | 3V3 |
| | 115 | VCCO_128 | 3V3 |
| | 116 | VCCO_129 | 3V3 |
| | 117 | VCCO_130 | 3V3 |
| | 118 | VCCO_131 | 3V3 |
| | 119 | VCCO_132 | 3V3 |
| | 120 | VCCO_133 | 3V3 |
| | 121 | VCCO_134 | 3V3 |
| | 122 | VCCO_135 | 3V3 |
| | 123 | VCCO_136 | 3V3 |
| | 124 | VCCO_137 | 3V3 |
| | 125 | VCCO_138 | 3V3 |
| | 126 | VCCO_139 | 3V3 |
| | 127 | VCCO_140 | 3V3 |
| | 128 | VCCO_141 | 3V3 |
| | 129 | VCCO_142 | 3V3 |
| | 130 | VCCO_143 | 3V3 |
| | 131 | VCCO_144 | 3V3 |
| | 132 | VCCO_145 | 3V3 |
| | 133 | VCCO_146 | 3V3 |
| | 134 | VCCO_147 | 3V3 |
| | 135 | VCCO_148 | 3V3 |
| | 136 | VCCO_149 | 3V3 |
| | 137 | VCCO_150 | 3V3 |
| | 138 | VCCO_151 | 3V3 |
| | 139 | VCCO_152 | 3V3 |
| | 140 | VCCO_153 | 3V3 |
| | 141 | VCCO_154 | 3V3 |
| | 142 | VCCO_155 | 3V3 |
| | 143 | VCCO_156 | 3V3 |
| | 144 | VCCO_157 | 3V3 |
| | 145 | VCCO_158 | 3V3 |
| | 146 | VCCO_159 | 3V3 |
| | 147 | VCCO_160 | 3V3 |
| | 148 | VCCO_161 | 3V3 |
| | 149 | VCCO_162 | 3V3 |
| | 150 | VCCO_163 | 3V3 |
| | 151 | VCCO_164 | 3V3 |
| | 152 | VCCO_165 | 3V3 |
| | 153 | VCCO_166 | 3V3 |
| | 154 | VCCO_167 | 3V3 |
| | 155 | VCCO_168 | 3V3 |
| | 156 | VCCO_169 | 3V3 |
| | 157 | VCCO_170 | 3V3 |
| | 158 | VCCO_171 | 3V3 |
| | 159 | VCCO_172 | 3V3 |
| | 160 | VCCO_173 | 3V3 |
| | 161 | VCCO_174 | 3V3 |
| | 162 | VCCO_175 | 3V3 |
| | 163 | VCCO_176 | 3V3 |
| | 164 | VCCO_177 | 3V3 |
| | 165 | VCCO_178 | 3V3 |
| | 166 | VCCO_179 | 3V3 |
| | 167 | VCCO_180 | 3V3 |
| | 168 | VCCO_181 | 3V3 |
| | 169 | VCCO_182 | 3V3 |
| | 170 | VCCO_183 | 3V3 |
| | 171 | VCCO_184 | 3V3 |
| | 172 | VCCO_185 | 3V3 |
| | 173 | VCCO_186 | 3V3 |
| | 174 | VCCO_187 | 3V3 |
| | 175 | VCCO_188 | 3V3 |
| | 176 | VCCO_189 | 3V3 |
| | 177 | VCCO_190 | 3V3 |
| | 178 | VCCO_191 | 3V3 |
| | 179 | VCCO_192 | 3V3 |
| | 180 | VCCO_193 | 3V3 |
| | 181 | VCCO_194 | 3V3 |
| | 182 | VCCO_195 | 3V3 |
| | 183 | VCCO_196 | 3V3 |
| | 184 | VCCO_197 | 3V3 |
| | 185 | VCCO_198 | 3V3 |
| | 186 | VCCO_199 | 3V3 |
| | 187 | VCCO_200 | 3V3 |
| | 188 | VCCO_201 | 3V3 |
| | 189 | VCCO_202 | 3V3 |
| | 190 | VCCO_203 | 3V3 |
| | 191 | VCCO_204 | 3V3 |
| | 192 | VCCO_205 | 3V3 |
| | 193 | VCCO_206 | 3V3 |
| | 194 | VCCO_207 | 3V3 |
| | 195 | VCCO_208 | 3V3 |
| | 196 | VCCO_209 | 3V3 |
| | 197 | VCCO_210 | 3V3 |
| | 198 | VCCO_211 | 3V3 |
| | 199 | VCCO_212 | 3V3 |
| | 200 | VCCO_213 | 3V3 |
| | 201 | VCCO_214 | 3V3 |
| | 202 | VCCO_215 | 3V3 |
| | 203 | VCCO_216 | 3V3 |
| | 204 | VCCO_217 | 3V3 |
| | 205 | VCCO_218 | 3V3 |
| | 206 | VCCO_219 | 3V3 |
| | 207 | VCCO_220 | 3V3 |
| | 208 | VCCO_221 | 3V3 |
| | 209 | VCCO_222 | 3V3 |
| | 210 | VCCO_223 | 3V3 |
| | 211 | VCCO_224 | 3V3 |
| | 212 | VCCO_225 | 3V3 |
| | 213 | VCCO_226 | 3V3 |
| | 214 | VCCO_227 | 3V3 |
| | 215 | VCCO_228 | 3V3 |
| | 216 | VCCO_229 | 3V3 |
| | 217 | VCCO_230 | 3V3 |
| | 218 | VCCO_231 | 3V3 |
| | 219 | VCCO_232 | 3V3 |
| | 220 | VCCO_233 | 3V3 |
| | 221 | VCCO_234 | 3V3 |
| | 222 | VCCO_235 | 3V3 |
| | 223 | VCCO_236 | 3V3 |
| | 224 | VCCO_237 | 3V3 |
| | 225 | VCCO_238 | 3V3 |
| | 226 | VCCO_239 | 3V3 |
| | 227 | VCCO_240 | 3V3 |
| | 228 | VCCO_241 | 3V3 |
| | 229 | VCCO_242 | 3V3 |
| | 230 | VCCO_243 | 3V3 |
| | 231 | VCCO_244 | 3V3 |
| | 232 | VCCO_245 | 3V3 |
| | 233 | VCCO_246 | 3V3 |
| | 234 | VCCO_247 | 3V3 |
| | 235 | VCCO_248 | 3V3 |
| | 236 | VCCO_249 | 3V3 |
| | 237 | VCCO_250 | 3V3 |
| | 238 | VCCO_251 | 3V3 |
| | 239 | VCCO_252 | 3V3 |
| | 240 | VCCO_253 | 3V3 |
| | 241 | VCCO_254 | 3V3 |
| | 242 | VCCO_255 | 3V3 |
| | 243 | VCCO_256 | 3V3 |
| | 244 | VCCO_257 | 3V3 |
| | 245 | VCCO_258 | 3V3 |
| | 246 | VCCO_259 | 3V3 |
| | 247 | VCCO_260 | 3V3 |
| | 248 | VCCO_261 | 3V3 |
| | 249 | VCCO_262 | 3V3 |
| | 250 | VCCO_263 | 3V3 |
| | 251 | VCCO_264 | 3V3 |
| | 252 | VCCO_265 | 3V3 |
| | 253 | VCCO_266 | 3V3 |
| | 254 | VCCO_267 | 3V3 |
| | 255 | VCCO_268 | 3V3 |
| | 256 | VCCO_269 | 3V3 |

U1004
ASE-50.000MHZ-LC-C-T

| | | | |
|---|---|---|---|
| 3V3 | 1 | VDD | |
| | 2 | GND | 0VDC_GND |
| 3V3 | 3 | STBY_N | |
| | 4 | OUT | FPGA_CLK_SOURCE |

U1005
SN74LVC1G08DCKT / RESET AND

| | | | |
|---|---|---|---|
| | 1 | NC | |
| FPGA_CRESET_N | 2 | A | |
| FPGA_SDRESET_RUN_N | 3 | B | |
| | 4 | Y | FPGA_LOCAL_RESET_N |
| | 5 | VCC | 3V3 |
| | 6 | GND | 0VDC_GND |

U1006
74LVC1G17GW / CDONE DELAY SCHMITT

| | | | |
|---|---|---|---|
| | 1 | NC | |
| FPGA_USER_RESET_RC | 2 | A | |
| | 3 | GND | 0VDC_GND |
| | 4 | Y | FPGA_USER_RESET_DELAYED |
| | 5 | VCC | 3V3 |

U1007
SN74LVC1G08DCKR / USER RESET AND

| | | | |
|---|---|---|---|
| | 1 | NC | |
| FPGA_USER_RESET_DELAYED | 2 | A | |
| FPGA_SDRESET_RUN_N | 3 | B | |
| | 4 | Y | FPGA_USER_RESET_N |
| | 5 | VCC | 3V3 |
| | 6 | GND | 0VDC_GND |

R1001 3V3 FPGA_CLK_SOURCE
R1002 3V3 FPGA_FLASH_CCK
R1003 3V3 FPGA_CDONE
R1004 3V3 FPGA_CFG_T0
R1005 3V3 FPGA_CFG_T0_N
R1006 3V3 FPGA_FLASH_CS_N
R1007 3V3 FPGA_CRAM_TDI
R1008 3V3 FPGA_CFG_CBSEL0
R1009 3V3 FPGA_CRAM_TCK
R1010 3V3 FPGA_CRAM_TMS
R1011 3V3 FPGA_CRAM_TDO
R1012 3V3 FPGA_CRAM_TMS
R1013 3V3 FPGA_CRAM_TDO
R1014 3V3 FPGA_FLASH_IO3
R1015 3V3 SDRESET_RUN_N
R1016 3V3 FPGA_CFG_WDRSVD
R1017 3V3 FPGA_CFG_CBSEL1
R1018 3V3 FPGA_CFG_CBSEL0
R1019 3V3 FPGA_CFG_CBSEL1
R1020 3V3 FPGA_USER_RESET_PC
R1021 3V3 FPGA_INDPOINT_READY
R1022 3V3 FPGA_MOTION_REQUEST
R1023 3V3 FPGA_PROGRESS_N
R1024 3V3 FPGA_CFG_CBSEL2
R1025 3V3 FPGA_USER_RESET_N
R1026 3V3 FPGA_LOCAL_RESET_N
R1027 3V3 FPGA_CFG_CBSEL3
R1028 3V3 FPGA_FLASH_IO2
R1029 3V3 FPGA_CLOCK_RESET_N

C1001 CORE_1V2 100nF 47V
C1002 CORE_1V2 100nF 47V
C1003 CORE_1V2 100nF 47V
C1004 CORE_1V2 100nF 47V
C1005 CORE_1V2 100nF 47V
C1006 CORE_1V2 100nF 47V
C1007 CORE_1V2 100nF 47V
C1008 CORE_1V2 100nF 47V
C1009 CORE_1V2 100nF 47V
C1010 3V3 100nF 47V
C1011 3V3 100nF 47V
C1012 3V3 100nF 47V
C1013 3V3 100nF 47V
C1014 3V3 100nF 47V
C1015 3V3 100nF 47V
C1016 3V3 100nF 47V
C1017 3V3 100nF 47V
C1018 3V3 100nF 47V
C1019 3V3 100nF 47V
C1020 3V3 100nF 47V
C1021 3V3 100nF 47V
C1022 3V3 100nF 47V
C1023 3V3 100nF 47V
C1024 3V3 100nF 47V
C1025 3V3 100nF 47V
C1026 3V3 100nF 47V
C1027 3V3 100nF 47V
C1028 3V3 100nF 47V
C1029 3V3 100nF 47V
C1030 3V3 100nF 47V
C1031 3V3 100nF 47V
C1032 3V3 100nF 47V
C1033 3V3 100nF 47V
C1034 3V3 100nF 47V
C1035 3V3 100nF 47V
C1036 3V3 100nF 47V
C1037 3V3 100nF 47V
C1038 3V3 100nF 47V
C1039 3V3 100nF 47V

J1 = J2 ARE INTERNAL LOGIC BOUNDARIES. NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESETN/74 plus two internal pullups screened. The separate 33k/1nF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial-power confirmation required.

J1001
POWER / QUALIFIED SEQUENCER

| | | | |
|---|---|---|---|
| CORE_1V2 | 1 | 1 | |
| 3V3 | 2 | 2 | |
| 0VDC_GND | 3 | 3 | |
| FPGA_CRESET_N | 4 | 4 | |
| FPGA_LOCAL_RESET_N | 5 | 5 | |
| FPGA_USER_RESET_N | 6 | 6 | |
| FPGA_CFG_WDRSVD | 7 | 7 | |
| 0VDC_GND | 8 | 8 | |

J1002
JTAG SERVICE / NO POWER INJECTION

| | | | |
|---|---|---|---|
| 3V3 | 1 | 1 | |
| 0VDC_GND | 2 | 2 | |
| FPGA_CRAM_TCK | 3 | 3 | |
| FPGA_CRAM_TMS | 4 | 4 | |
| FPGA_CRAM_TDI | 5 | 5 | |
| FPGA_CRAM_TDO | 6 | 6 | |
| FPGA_CDONE | 7 | 7 | |
| | 8 | 8 | |

J1003
HOST SPI / 3V3 ONLY

| | | | |
|---|---|---|---|
| HOST_SPI_CLK | 1 | 1 | |
| HOST_SPI_CS_N | 2 | 2 | |
| HOST_SPI_MOSI | 3 | 3 | |
| HOST_SPI_MISO | 4 | 4 | |
| 0VDC_GND | 5 | 5 | |
| FPGA_USER_RESET | 6 | 6 | |
| FPGA_WATCHDOG_FAULT | 7 | 7 | |
| FPGA_TIMING_FAULT | 8 | 8 | |

J1004
HW PERMISSION / NO ESP MOTION AUTHORITY

| | | | |
|---|---|---|---|
| RAW_ANYBRT_N | 1 | 1 | |
| FPGA_ESMSTPOFF_ACKNT | 2 | 2 | |
| FPGA_MOTION_REQUEST | 3 | 3 | |
| FPGA_PROGRESS_N | 4 | 4 | |
| 3V3_PUMP | 5 | 5 | |
| FPGA_ESMSTPOFF_REQ | 6 | 6 | |
| FPGA_LOCAL_RESET_N | 7 | 7 | |
| 0VDC_GND | 8 | 8 | |

J1005
TWELVE MOTOR STEP / DIR

| | | | |
|---|---|---|---|
| STEP0_FPGA | 1 | 1 | |
| DIR0_FPGA | 2 | 2 | |
| STEP1_FPGA | 3 | 3 | |
| DIR1_FPGA | 4 | 4 | |
| STEP2_FPGA | 5 | 5 | |
| DIR2_FPGA | 6 | 6 | |
| STEP3_FPGA | 7 | 7 | |
| DIR3_FPGA | 8 | 8 | |
| STEP4_FPGA | 9 | 9 | |
| DIR4_FPGA | 10 | 10 | |
| STEP5_FPGA | 11 | 11 | |
| DIR5_FPGA | 12 | 12 | |
| STEP6_FPGA | 13 | 13 | |
| DIR6_FPGA | 14 | 14 | |
| STEP7_FPGA | 15 | 15 | |
| DIR7_FPGA | 16 | 16 | |
| STEP8_FPGA | 17 | 17 | |
| DIR8_FPGA | 18 | 18 | |
| STEP9_FPGA | 19 | 19 | |
| DIR9_FPGA | 20 | 20 | |
| STEP10_FPGA | 21 | 21 | |
| DIR10_FPGA | 22 | 22 | |
| STEP11_FPGA | 23 | 23 | |
| DIR11_FPGA | 24 | 24 | |

J1006
ENABLE RESERVES / CIRCUITS ABSENT

| | | | |
|---|---|---|---|
| ENA_N_FPGA | 1 | 1 | |
| ENA_N_FPGA | 2 | 2 | |
| ENA_N_FPGA | 3 | 3 | |
| ENA_N_FPGA | 4 | 4 | |
| ENA_N_FPGA | 5 | 5 | |
| ENA_N_FPGA | 6 | 6 | |
| ENA_N_FPGA | 7 | 7 | |
| ENA_N_FPGA | 8 | 8 | |
| ENA_N_FPGA | 9 | 9 | |
| ENA_N_FPGA | 10 | 10 | |
| ENA_N_FPGA | 11 | 11 | |

MC LOCAL POWER — IS-2 HARDWARE PROPOSAL / NOT MANUFACTURING READY

CORE_1V2: 1.2 V / 0.8 A reservation; 1 MHz forced PWM

IO_3V3: 3.3 V / 0.15 A reservation; 2.5 MHz forced PWM

U2001
TPS62901RFJR
C2001
10uF 25V X7R 10%
L2001
2.2uH
C2002
22uF 10V X7S 20%
R2001
76.8k 1%
R2002
10k 1%
C2003
6.8nF 50V C0G 5%
R2003
100k 1%
R2021
97.6k 0.01% 5ppm
R2029
92.1k 0.01% 5ppm
C2007
100nF X7R 100V 10%
U2002
TPS62901RFJR
C2004
10uF 25V X7R 10%
L2002
1uH
C2005
22uF 10V X7S 20%
R2005
32.4k 1%
R2006
10k 1%
C2006
6.8nF 50V C0G 5%
R2007
100k 1%
R2008
301k 1%
R2041
1.87k 0.01% 5ppm
R2049
1.15k 0.01% 5ppm
C2008
100nF X7R 100V 10%
R2061
94.3k 0.01% 5ppm
R2069
57.6k 0.01% 5ppm
C2009
100nF X7R 100V 10%
U2003
LTC2962HUD#TRPBF
R2042
261 0.01% 5ppm
R2050
316 0.01% 5ppm
R2051
4.02 1% 100ppm
U2004
LTC2962HUD#TRPBF
U2005
LTC2962HUD#TRPBF
R2027
10k 0.01% 5ppm
R2035
10k 0.01% 5ppm
R2047
1k 0.01% 5ppm
R2055
1k 0.01% 5ppm
R2067
10k 0.01% 5ppm
R2075
10k 0.01% 5ppm
R2101
3.3k 1%
R2102
3.3k 1%
R2103
10k 1%
R2104
6.8k 1%
R2105
10k 1%
R2106
47k 1%
R2107
470k 1%
U2006
SN74LVC1G17DBVR
R2108
4.7k 1%
C2010
100nF X7R 100V 10%
R2001
MC POWER REVIEW BOUNDARY

0 V window releases core; core window releases IO; IO window releases FPGA reset.
Each supervisor releases 160–240 ms. Raw MC_RAILS_OK_5V is not a 3.3 V signal.
FPGA_CRESET_N is the buffered rails-good term; combine other reset requests upstream.
Regulator PG pins are diagnostic only; no PG-to-EN cascade. Physical connector is external.

Actual rail limits retained: 5 V 4.85–5.15; T20 core 1.15–1.25; IO 3.20–3.40 V.
Core thresholds include initial/temp plus 400ppm drift; historical tight ledger is distinct.
IO uses 2.5 MHz because existing FPGA decoupling exceeds the 1 MHz capacitance limit.
Proposed MC sum: 0.50 A steady / 0.80 A downstream startup input charging separate.
Arbitrary partial power, assembled stress, brownout slew, shutdown and timing unproved.

Sheet: /mc_local_power/
File: mc_local_power.kicad_sch
Title: MC local regulators and independent window sequencing
Size: A1 | Date: 2026-09-30 | Rev: r0.1 / IS-2 proposal
KiCad E.D.A. 10.0.4 | Id: 3/6

# IS-2 PROPOSAL: DUAL WATCHDOG STARTUP CONDITIONING — SEPARATE REVIEW CANDIDATE

NOT CONNECTED TO THE ASSEMBLY. No complete permission chain, latches, field outputs, PCB or physical qualification.

Separate timers / shared IO_3V3 and LOGIC_GND / G1B threshold is not a 3.3V power-good qualification / all inputs INTERNAL logic.

4.7k FPGA-facing pulls: SET1 = request AND FPGA_USER_RESET_N. Active-HIGH polarity / 1us pulse contract unchanged. MR open.

WDI: one 1us high pulse, idle LOW. FALLING edges feed timer.
Old toggle-per-UPDATE gateware is INCOMPATIBLE.
3.3nF COG calculated timing, never a measured2ms proof.

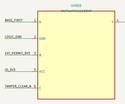
U3001
TPS3851G30DSWRQ1
WD_M_CWD
WD_M_SET1_BUF
WD_M_WDI_BUF
WDO_N
RESET_N
WD_M_HEALTH_RAW
FPGA_MOTION_REQUEST
R3003
4.7k 1%
C3001
3.3nF C0G 50V 5%
IO_3V3
R3001
10k 1%
U3002
TPS3851G30DSWRQ1
WD_O_CWD
WD_O_SET1_BUF
WD_O_WDI_BUF
WD_O_HEALTH_RAW
FPGA_OUTPUT_REQUEST
R3004
4.7k 1%
C3002
3.3nF C0G 50V 5%
R3002
10k 1%
U3003
SN74LVC1G34DBVR
FPGA_PROGRESS_M
WD_M_WDI_BUF
FPGA_PROGRESS_M
R3005
4.7k 1%
R3006
100k 1%
U3004
SN74LVC1G34DBVR
FPGA_PROGRESS_O
WD_O_WDI_BUF
FPGA_PROGRESS_O
R3007
4.7k 1%
R3008
100k 1%
Each RESET = WDO wired-AND with
10k pull-up. WD_M_OK / WD_O_OK
have100k pulls: partial power UNQUALIFIED.
No sticky permission latch in this block.
U3005
SN74LVC2G17DBVR
WD_M_HEALTH_RAW
WD_O_HEALTH_RAW
WD_M_OK
WD_O_OK
4.7k replaces four FPGA-facing100k pulls.
Two SET1 AND gates add reset guard / LOW drive.
Two WDI gates add toggle guard / LOW drive.
SET1 pin leakage has NO published maximum.
Loaded LOW, reset slew and partial power UNQUALIFIED.
WD_M_OK
R3009
100k 1%
WD_O_OK
R3010
100k 1%
U3006
SN74LVC1G08DCKR
FPGA_MOTION_REQUEST
FPGA_USER_RESET_N
WD_M_SET1_BUF
U3007
SN74LVC1G08DCKR
FPGA_OUTPUT_REQUEST
FPGA_USER_RESET_N
WD_O_SET1_BUF
WD_M_SET1_BUF
R3011
100k 1%
WD_O_SET1_BUF
R3012
100k 1%
B3001
INTERNAL REVIEW BOUNDARY
FPGA_USER_RESET_N
R3013
4.7k 1%
IO_3V3
LOGIC_GND
FPGA_MOTION_REQUEST
FPGA_OUTPUT_REQUEST
FPGA_PROGRESS_M
FPGA_PROGRESS_O
WD_M_OK
WD_O_OK
FPGA_USER_RESET_N
C3003
100nF X7R 100V 10%
C3004
100nF X7R 100V 10%
C3005
100nF X7R 100V 10%
C3006
100nF X7R 100V 10%
C3007
100nF X7R 100V 10%
C3008
100nF X7R 100V 10%
C3009
100nF X7R 100V 10%
LOGIC_GND

DRB copper-pad candidate only: paste/stencil, thermal vias, CWD guarding/cleanliness/leakage, routing, partial power and brownout remain unqualified.

Sheet: /watchdog_conditioning/
File: watchdog_conditioning.kicad_sch
Title: IS-2 proposed watchdog conditioning — review only
Size: A1 | Date: 2026-09-30 | Rev: r0.2-proposal
KiCad E.D.A. 10.0.4 | Id: 4/6

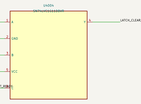
MC PERMISSION / STICKY INTERRUPTION CORE — UPSTREAM QUALIFICATION AND FIELD DRIVERS REMAIN SEPARATE



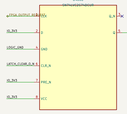






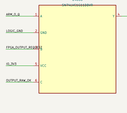


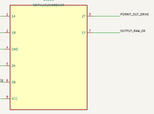


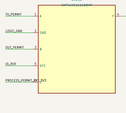








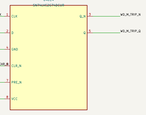


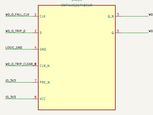


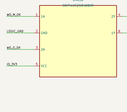






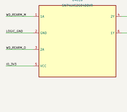




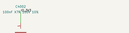






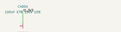




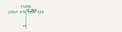


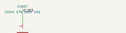


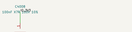


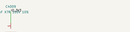


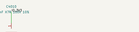




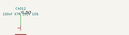



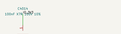






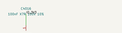


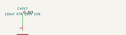


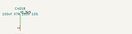


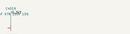


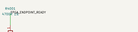




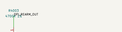



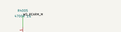








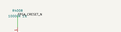


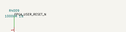


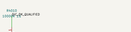


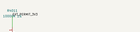












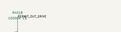






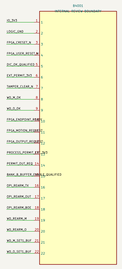


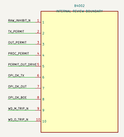









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File: mc_permission.kicad_sch

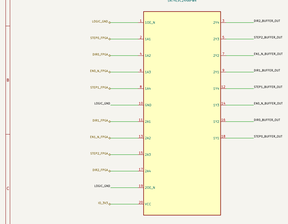
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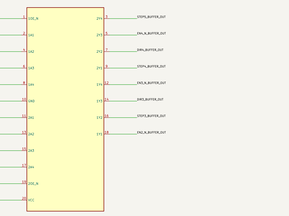
Size: A0 | Date: 2026-09-30 | Rev: r0.2

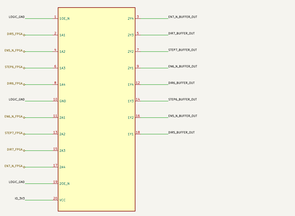
KiCad 8.0.4 10.0.4 | Id: 5/6

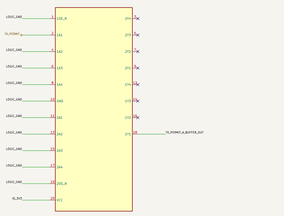
No manufacture/physical timing claim. IS-1 arm/DPI equations; N-TX GPIO startup pulls need proposed IS-2 reconciliation. WD_TRIP_N = WD_M_TRIP_N | WD_EL_TRIP_N. Capture actual timer SETS/ RESETS, feedback, power-reset or explicit WD_BEAMN pulse clears the record. BG/BQ are logical boundaries, not connectors. Trip clear protocol, loaded edge timing and upstream qualified terms remain open.

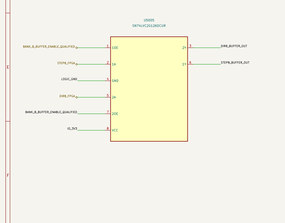
MC MOTOR / TX Ioff: 8 STANDARD +4 EXPANSION; B1–B4 ARE LOGICAL BOUNDARIES, NOT CONNECTORS

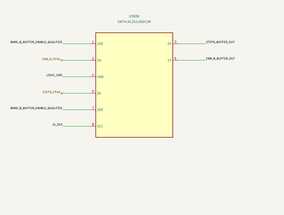


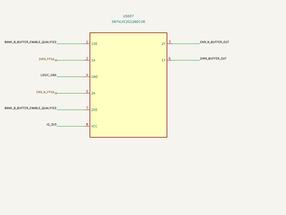



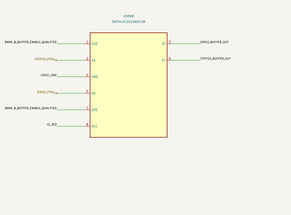



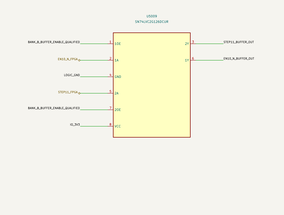



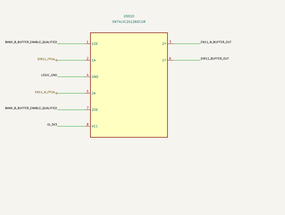



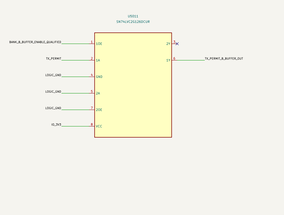



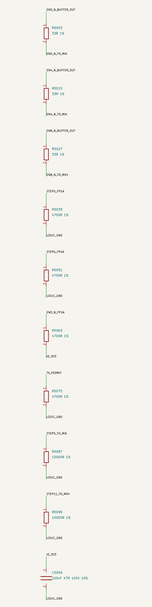



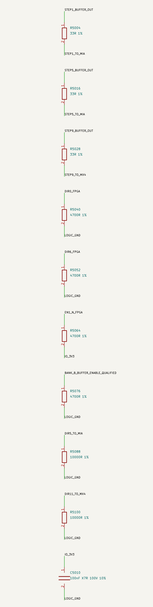

















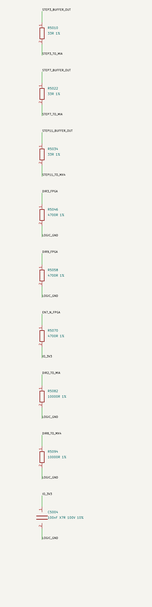








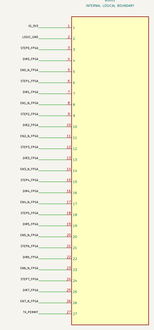


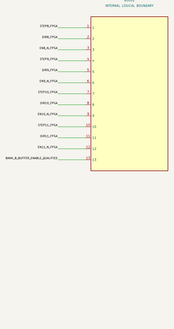



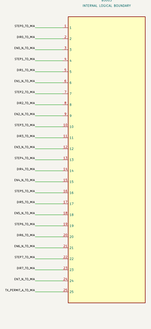


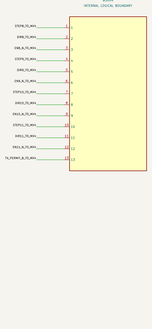





Bank0/Y44 OEs tied LOW(active). Bank0B2G126 used OEs share BANK_B_BUFFER_ENABLE_@UNUSED with parentOPI_BOI rinet.
TX_N outgoing pull-ups belong on powered/MCU/MAX receiver rails: B2V5B on sheet4; rail, 330 series are unqualified SI candidates.
Not complete/RC no routing, tangential, loadedtiming or manufacture claim.

Sheet: /mc_motor_buffers/
File: mc_motor_buffers.kicad_sch

**Title: Motion-card motor Ioff boundaries**

| | | |
|---|---|---|
| Size: A0 | Date: 2026-09-30 | Rev: r0.1 |
| KiCad 10.0.4 | | Id: 6/9 |